

Building Thinking Classrooms Look For Guide

Implementation of the 14 Practices of Building Thinking Classrooms

The Building Thinking Classrooms (BTC) framework by Peter Liljedahl serves as a transformative guide for fostering student-centered classrooms that prioritize critical thinking and collaboration. The goal of BTC implementation is to help educators create environments where students are actively engaged in sustained problem-solving and meaningful discourse. This "Look-For" document provides districts and schools with a scaffolded approach to implementation, organized into four toolkits, each focusing on distinct practices that gradually build toward these outcomes.

This document outlines observable behaviors and evidence for leaders to assess progress while providing educators with clear goals to target at each stage of BTC implementation. Schools and districts can use this scaffolded approach to meet teachers where they are in their professional learning journey, empowering them to gradually adopt practices that lead to deeper student learning and engagement.

Purpose of the Document

The purpose of this document is threefold:

1. **Support Gradual Implementation:** By organizing BTC practices into toolkits, schools can focus on foundational practices (Toolkit 1) before moving to more advanced practices (Toolkit 2-4). This progression ensures that teachers have the time and resources needed to master each phase.
2. **Provide Clear Indicators:** Observable educator and student behaviors, as well as evidence aligned with the intended outcomes, help educators and leaders monitor and evaluate progress.
3. **Encourage Reflection:** The alignment between intended outcomes and practices encourages educators to reflect on their teaching methods, ensuring alignment with the overarching goals of critical thinking and collaboration.

Focus on Intended Outcomes

The practices outlined in each toolkit are designed to build capacity in both educators and students to meet the following intended outcomes:

1. **Student-Centered Classrooms:** Through non-traditional tasks, random grouping, and collaborative methods, educators shift from traditional instruction to environments where students take ownership of their learning.
2. **Sustained Engagement and Problem-Solving:** BTC encourages students to embrace productive struggle, engage with peers in meaningful discussions, and persist through challenging tasks, leading to deeper and more sustained engagement in their learning.

Each toolkit represents a step toward achieving these outcomes, beginning with the foundational strategies of Toolkit 1 and progressing to advanced assessment practices in Toolkit 4. The observable behaviors and evidence provided in this document offer leaders a lens to measure the success of BTC implementation and ensure alignment with the district's vision for teaching and learning.

Toolkit 1: Building a Culture of Thinking

The first step in implementing the Building Thinking Classrooms framework is establishing a foundation where students engage in meaningful collaboration and critical thinking. Toolkit 1 focuses on creating this environment by introducing foundational practices that encourage active participation and visible thinking. These practices include:

- Begin lessons with good problem-solving tasks: Tasks that spark curiosity and promote deeper thinking.
- Form visibly random groups: Ensure diverse collaboration and equal participation.
- Use vertical non-permanent surfaces (VNPS): Encourage standing, movement, and visible student thinking.
- Provide oral instructions: Engage students immediately and reduce over-reliance on written instructions.

The Begin Lessons with Good Problem-Solving Tasks foundational practices not only help establish a collaborative culture but also create opportunities for students to actively engage in problem-solving and group discussions. Below are the specific "look-fors" that help educators and leaders measure progress toward the intended outcomes.

Practice	Observable Educator Behaviors	Observable Student Behaviors	Evidence Leaders May See
Begin lessons with good problem-solving tasks	• Selects tasks that are accessible (low-floor) and challenging (high-ceiling). • Introduces tasks that connect to students' prior knowledge but require new thinking. • Provides clear context and purpose for tasks.	• Actively engages in tasks with curiosity and enthusiasm. • Uses prior knowledge and experiments with novel strategies. • Discusses and connects the task to its purpose or real-world relevance.	• Lesson plans featuring rich, open-ended tasks. • Examples of tasks completed by students showing diverse approaches. • Observation notes of students engaging with tasks immediately after instruction.
Form Visibly Random Groups	• Uses methods (e.g., cards, names, or apps) to randomly group students daily.	• Interacts with and collaborates with peers they may not usually work with. • Builds trust and relationships	• Logs or documentation of random grouping methods used regularly. • Positive group dynamics

	- Explains the purpose of random grouping to promote equity and collaboration. - Rotates groups frequently to ensure consistent variety.	with different peers over time. Adapts quickly to new group configurations and engages collaboratively.	observed during classroom walkthroughs.
Use Vertical Non-Permanent Surfaces (VNPS)	- Sets up VNPS (e.g., whiteboards, posters, windows) for group work. - Encourages all group members to contribute to the VNPS. - Monitors and facilitates group work while allowing independence.	- Stands and actively contributes ideas to the VNPS. - Writes, draws, or explains their thinking on VNPS. - Revisits and refines their work collaboratively.	- Classroom setups with visible evidence of VNPS use. - Observation notes or photos capturing student collaboration at VNPS. - Progression of ideas documented on VNPS during walkthroughs.
Provide Oral Instructions	- Gives brief, clear instructions orally without writing them down. - Repeats or rephrases instructions as needed for clarity. - Stays accessible to answer process-related questions after instruction.	- Listens attentively and begins tasks immediately. - Asks clarifying questions to ensure understanding. - Works independently but seeks clarification when necessary.	- Observation notes of students engaging quickly after receiving oral instructions. - Reduced instances of students appearing confused or disengaged during task initiation. - Feedback from students showing they understand the expectations without over-reliance on the teacher.

Toolkit 2: Nurturing Student Autonomy

Building on the foundational practices of Toolkit 1, Toolkit 2 shifts the focus toward fostering greater student independence. This toolkit empowers students to take ownership of their learning by encouraging self-reliance, persistence, and deeper engagement with challenging tasks. The practices in this toolkit include:

- **Answer only keep-thinking questions:** Guide students to think critically and persist in solving problems rather than providing direct answers.
- **Give students autonomy to access answers and resources:** Allow students to seek out tools and materials independently, fostering resourcefulness and self-directed learning.
- **Use hints and extensions to manage flow:** Keep students engaged by offering targeted support when necessary and providing extensions to challenge advanced learners.

These practices nurture a classroom environment where students are encouraged to embrace productive struggle, develop confidence in their ability to solve problems, and take initiative in their learning. Below are the specific "look-fors" that help educators and leaders assess the successful implementation of these strategies.

Intended Learning Outcome	Observable Educator Behaviors	Observable Student Behaviors	Evidence Leaders May See
Answer Only Keep-Thinking Questions	● Responds to student questions with prompts or guidance that encourage deeper thinking. ● Redirects students to their peers or resources for further exploration instead of providing direct answers. ● Models curiosity and inquiry by asking clarifying	● Poses thoughtful questions to clarify understanding or expand thinking. ● Perseveres through challenges, seeking answers through exploration and collaboration. ● Engages in discussions with peers to brainstorm and refine ideas.	● Observation notes showing educators redirecting student questions toward thinking processes. ● Examples of student collaboration to solve problems without reliance on the teacher. ● Feedback or reflections from students highlighting increased confidence in their

	or leading questions.		problem-solving abilities.
Give Students Autonomy to Access Answers and Resources	- Creates a classroom environment where students can independently access tools, materials, or resources. - Clearly communicates expectations for using resources and seeking answers. - Encourages students to validate their own solutions using available tools or references.	- Independently seeks out resources to find answers or solve problems. - Demonstrates self-regulation in deciding when and how to use tools or materials. - Uses resources collaboratively with peers to confirm or challenge their thinking.	- Classroom setups with readily accessible materials and resources for students. - Student work showing use of tools, charts, or references to enhance problem-solving. - Observations of students navigating and using resources effectively without constant teacher intervention.
Use Hints and Extensions to Manage Flow	- Provides hints to guide students toward solutions without giving away answers. - Differentiates support by offering extensions for students who are ready for additional challenges. - Actively monitors group progress to adjust pacing and support as needed.	- Uses hints to advance their thinking and overcome obstacles. - Engages in extended tasks or challenges to deepen understanding when ready. - Maintains focus and persistence, adapting their approach based on teacher feedback or scaffolding.	- Observation notes capturing the use of hints to scaffold student learning. - Student work reflecting the use of extensions to explore more complex ideas. - Differentiated task materials or examples showing varying levels of student engagement with hints and challenges.

Toolkit 3: Consolidating Thinking

Toolkit 3 focuses on solidifying student understanding by helping them organize and reflect on their learning. This stage emphasizes collaborative consolidation of ideas and encourages students to take ownership of the knowledge they've built. The practices in this toolkit include:

- **Leveling to the bottom:** Bring the class together to ensure all students have a shared foundational understanding before progressing.
- **Use student work to build notes:** Highlight student-generated work as examples to reinforce key concepts and create a sense of ownership.
- **Engage in meaningful note-making:** Guide students to create notes that reflect their thinking process and deepen their comprehension.

These practices ensure that students actively engage in organizing and internalizing their learning while collaborating with peers to consolidate ideas. Below are the specific "look-fors" that help educators and leaders measure the successful implementation of these strategies.

Intended Learning Outcome	Observable Educator Behaviors	Observable Student Behaviors	Evidence Leaders May See
Leveling to the Bottom	● Monitors group progress and ensures all students reach a foundational understanding before moving forward. ● Brings the class together to discuss and solidify key concepts once a baseline level of understanding is achieved. ● Uses clarifying questions and examples to bridge gaps in understanding	● Engages in discussions to consolidate foundational ideas. ● Demonstrates a willingness to revisit concepts to ensure a shared understanding. ● Collaborates to help peers who may need additional support to reach the same level of understanding. ●	● Observation notes of whole-class discussions focused on foundational ideas after group work. ● Documentation of common understandings reached during classroom consolidation sessions. ● Student work reflecting a shared baseline of comprehension before advancing.

	across groups.		
Use Student Work to Build Notes	• Selects and highlights student work as exemplars during class discussions. • Facilitates a process where students contribute their ideas and strategies to collective notes. • Frames student contributions as valuable to the learning process, emphasizing ownership.	• Shares their work or strategies confidently with the class. • Analyzes peer contributions to identify key ideas and strategies. • Actively participates in synthesizing class notes based on group discussions and shared work.	• Samples of student work used during class discussions to build notes. • Observation notes capturing moments when students' ideas are incorporated into collective understanding. • Finalized notes that clearly reflect diverse student contributions.
Engage in Meaningful Note-Making	• Guides students in creating notes that capture key concepts, strategies, and processes. • Encourages students to use their own language, diagrams, or visual representations in their notes. • Models the value of meaningful notes as a tool for reflection and future reference.	• Creates personal, detailed, and organized notes during or after class discussions. • Engages in reviewing and revising notes to reflect deeper understanding over time. • Uses notes as a resource to solve problems or review concepts.	• Samples of student notes showing depth and personalization. • Observation notes of students actively engaged in the note-making process. • Evidence of students using notes during independent or group work.

Toolkit 4: Assessing and Evaluating Thinking

Toolkit 4 emphasizes assessment practices that prioritize the thinking process over simple correctness. By focusing on formative and summative assessments that value critical thinking, reflection, and feedback, this toolkit ensures that students are not only learning content but also developing their ability to think deeply and independently. The practices in this toolkit include:

- **Use formative assessment to inform instruction:** Continuously gather evidence of student understanding to adapt teaching strategies in real time.
- **Provide feedback that promotes thinking:** Offer feedback that encourages students to refine their ideas and engage in deeper inquiry.
- **Encourage self-assessment and reflection:** Empower students to evaluate their own learning and identify areas for growth.
- **Design summative assessments that value thinking processes:** Develop assessments that reward the process of problem-solving and reasoning, not just the final answer.

These practices ensure that assessment becomes a meaningful part of the learning process, helping students grow as thinkers while providing teachers with actionable insights. Below are the specific "look-fors" that help educators and leaders evaluate the successful implementation of these strategies.

Intended Learning Outcome	Observable Educator Behaviors	Observable Student Behaviors	Evidence Leaders May See
Use Formative Assessment to Inform Instruction	● Regularly checks for understanding during lessons using questioning, observation, or quick tasks. ● Adjusts instruction or scaffolding based on real-time student needs and progress. ● Uses formative assessments to identify	● Responds to informal checks for understanding, such as quick quizzes, verbal questions, or exit tickets. ● Demonstrates improved understanding or adjustments based on teacher feedback. ● Engages in clarifying misunderstandings when prompted by formative	● Examples of formative assessments, such as exit tickets or mini-assessments, with feedback. ● Observation notes showing how teachers adjust instruction based on student responses. ● Evidence of increased student comprehension or reduced misconceptions after formative

	trends or misconceptions for targeted interventions.	assessments.	interventions.
Provide Feedback That Promotes Thinking	• Delivers feedback that focuses on thinking processes rather than just correctness. • Encourages students to revisit their work and refine their ideas based on feedback. • Models curiosity and inquiry through feedback that poses further questions or challenges.	• Revises work based on feedback to deepen understanding. • Explains how they incorporated feedback into their thinking or solutions. • Demonstrates increased confidence and independence in approaching similar problems.	• Student work showing revisions or improvements based on feedback. • Observation notes capturing moments when feedback sparks further student thinking. • Feedback documentation, such as written or verbal notes, highlighting specific thinking-focused suggestions.
Encourage Self-Assessment and Reflection	• Provides tools or prompts for students to evaluate their own learning and thinking processes. • Facilitates structured reflection opportunities, such as journals, discussions, or checklists. • Models reflective practices by sharing examples of how to analyze one's learning journey.	• Uses self-assessment tools to identify strengths and areas for improvement. • Reflects on their thinking and problem-solving strategies through written or verbal communication. • Sets goals for their learning based on self-reflection and assessment.	• Student reflections, such as journal entries or self-assessment rubrics. • Observation notes showing students engaging in metacognitive discussions about their learning. • Growth in students' ability to articulate their strengths and challenges over time.
Design Summative	• Develops assessments	• Demonstrates reasoning and	• Summative assessment tasks

Assessments That Value Thinking Processes	that prioritize critical thinking, problem-solving, and process over rote answers. • Includes tasks that allow for multiple solution pathways or creative approaches. • Uses rubrics that assess reasoning, strategy, and explanation, not just final answers.	explains their thinking in assessments. • Engages with assessments as an opportunity to showcase their process and understanding. • Approaches complex tasks with confidence and creativity.	featuring open-ended problems or real-world applications. • Rubrics that highlight thinking and process-based criteria. • Student work showing detailed explanations, varied approaches, and evidence of critical thinking.
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Conclusion

The **Building Thinking Classrooms** framework provides a powerful, phased approach to transforming classrooms into spaces that prioritize critical thinking, collaboration, and student autonomy. By implementing the practices outlined in Toolkits 1-4, educators progressively build a culture of thinking, nurture independence, consolidate understanding, and refine assessment strategies that value the thinking process. Each toolkit contributes to achieving the intended outcomes: fostering student-centered classrooms and cultivating sustained engagement and problem-solving skills. This "Look-For" document serves as a guide for educators and leaders to monitor progress, reflect on practices, and celebrate successes along the journey. With a clear focus on observable behaviors and evidence, schools can ensure that the BTC framework is implemented effectively, resulting in meaningful and lasting improvements in teaching and learning. Together, these practices create a pathway toward empowering all students to become confident, critical thinkers prepared for the challenges of the future.